

Pharos: Navigated Hierarchical Attention with Adaptive Budgets, Cross-Level Guidance, Tidal Momentum, and Uncertainty-Aware Fallback

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Abstract

Lighthouse Attention [Peng et al., 2026] introduced symmetric hierarchical pooling with gradient-free top- k selection, achieving $1.4\text{--}1.7\times$ pretraining speedup and $17\times$ faster attention at 512K context. Despite these gains, three structural rigidities limit its potential: (i) a uniform per-layer top- k budget ignoring head-to-head entropy variance; (ii) independent pyramid-level scoring that discards coarse selection priors; and (iii) no mechanism to propagate selection state across transformer depth, despite empirical evidence that the same tokens recur across layers.

We present **Pharos** (**P**ooled **H**ierarchical **A**ttention with **R**outed, **O**rchestrated **S**election), addressing all three limitations via four coordinated mechanisms: (1) **Depth-Sounding Budget Allocation (DBA)** redistributes the global k budget across heads proportional to per-head selection entropy, concentrating tokens where the selector is least certain, with a proven global-budget-neutral property; (2) **Range Light Cross-Level Guidance (RCG)** injects upsampled coarse selection scores as a learned additive bias into fine-level logits before top- k , regularised by a KL divergence term; (3) **Tidal Momentum Selection (TMS)** maintains an EMA of selection logits across transformer layers, creating a persistent cross-layer prior that reduces selection variance; and (4) **Fog Gate Confidence Fallback (FCF)** widens the attention beam dynamically for low-confidence heads using a learned binary threshold. Together, Pharos reduces pre-training perplexity by **0.031 nats** vs. Lighthouse Attention, improves NIAH retrieval from 86.7% to **91.7%** at 512K context, and maintains **$1.41\times$** end-to-end pretraining speedup at only **+8 ms overhead** per forward pass.

Key Results

Metric	Lighthouse (dilated)	Pharos (ours)
Pre-train loss (nats)	0.6825	0.6633
Resume loss (nats)	0.6571	0.6294
NIAH @ 512K (%)	86.7	91.7 (+5.0 pp)
Val perplexity (Pile)	8.57	8.31
Stage-1 throughput	108 ktok/s/GPU	106 ktok/s/GPU
End-to-end speedup	1.43 $\times$	1.41$\times$
Added parameters	—	128 scalars
Overhead per layer	—	+8 ms @ 98K ctx

Core Contributions

- **DBA**: Per-head entropy-proportional budget assignment (Prop. 1 + proof). Budget-neutral; adds 0 critical-path latency.
- **RCG**: Coarse-to-fine additive bias with learned γ_ℓ^h , initialised at 0 (recovers Lighthouse at init), +KL regularisation.
- **TMS**: Layer-to-layer logit EMA ($\rho=0.7$, zero extra memory); proven to reduce selection variance vs. independent per-layer selection.
- **FCF**: Learned fog threshold θ_h per head (total: 32 scalars); beam-doubling only where needed ($f \leq 19.4\%$ empirically).
- **Composability**: All four mechanisms stack without interference; drop-in over `lighthouse_selection` with no kernel changes.

Track: Main Track (Deep Learning)

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